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# Self-Supervised Representation Learning on STL-10: An Empirical Study of SimCLR and Barlow Twins

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Reza Jahani {rjahani@ncsu.edu}<sup>1</sup>

## Abstract

We study self-supervised representation learning (SSL) on the STL-10 dataset by implementing and empirically comparing two popular methods among SSL methods: SimCLR, a contrastive approach based on instance discrimination via the InfoNCE objective, and Barlow Twins, a non-contrastive method that learns representations by driving the cross-correlation matrix of two augmented views toward the identity. Both models share the same convolutional backbone with a projection head, and are pretrained on the unlabeled split of STL-10. We further implement a supervised baseline in which the same feature extractor backbone is adopted for neural architecture and is trained end-to-end on the labeled training set of STL-10. After pretraining, the representations are evaluated via a linear probe trained on increasing fractions of the labeled set (10 to 500 images per class). Our results validate the vital benefit of SSL: both methods learn transferable representations that approach the supervised baseline, with SimCLR reaching 62.49% and Barlow Twins reaching 59.46% at 500 labels per class, compared to 60.35% for the supervised baseline. In the very low-label regime, the self-supervised representations can effectively replace supervised learning, empirically confirming that self-supervised pretraining is an effective strategy when labeled data is not available or desirable.

## 1. Introduction

Modern deep learning pipelines rely heavily on large volumes of labeled data to achieve strong performance on visual tasks. Manual annotation or label collection, particularly for prediction tasks such as semantic segmentation or detection, can come at high cost. This practical bottleneck has motivated a long line of work on learning useful visual representations from unlabeled data, a paradigm broadly referred to as self-supervised learning (SSL). The central premise of SSL is that the structure of the data itself, when paired with an appropriately designed pretext task, carries

sufficient signal to learn features that transfer well to downstream tasks. Once such features and representations are learned, they can be adapted to a target task with a small amount of labeled data, often via a simple linear probe or finetuning.

Among the many families of SSL methods proposed in recent years, contrastive learning has emerged as an effective and conceptually simple approach. The main intuition of these methods is that two different augmented views of the same image should be mapped close together in feature space, while views of different images should be pushed apart. The SimCLR framework (Chen et al., 2020a;b), building on earlier work on contrastive predictive coding (Oord et al., 2018) and recent advances in supervised contrastive learning (Khosla et al., 2020), formalized this idea in a simple and effective manner using the InfoNCE objective. In parallel, a second family of methods has avoided the explicit contrastive formulation altogether, relying instead on architectural tricks or information-theoretic objectives to prevent representational collapse. BYOL (Grill et al., 2020) accomplishes this through a momentum-updated teacher network and asymmetric predictors, while Barlow Twins (Zbontar et al., 2021) takes a notably different route: it directly regularizes the cross-correlation matrix of the two views’ embeddings toward the identity, simultaneously enforcing view invariance and feature decorrelation.

The quality of the representations learned by such methods heavily depends on the choice of data augmentations. Composition of random cropping and color distortion has been identified as especially important, since stronger augmentations force the encoder to rely on high-level semantic content rather than low-level statistical shortcuts such as color histograms or texture patches. The STL-10 dataset (Chen et al., 2020a), with its 100000 unlabeled images and only 500 labeled images per class, is particularly well suited for studying this regime. Figure 1 shows a few examples of common augmentations widely used in SSL.

In this project, we implement, train, and empirically evaluate two SSL methods on STL-10: SimCLR as a representative contrastive method, and Barlow Twins as a representative redundancy-reduction method. For a fair comparison, both methods share the same convolutional backbone, projection

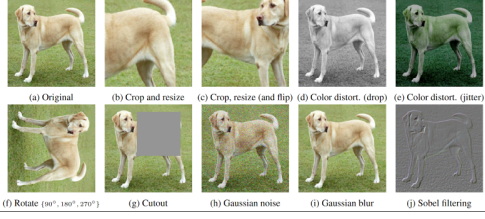


Figure 1. Examples of data augmentations helpful for SSL.

head dimensionality, and training budget. As a supervised baseline, we additionally train the same backbone on the labeled training split of STL-10. Evaluation is performed via a linear probe on features extracted from the frozen backbones, across a range of labeled samples per class. Our main works in this project is summarized as follows:

- We implement simCLR and BarlowTwins methods as two SSL methodologies under similar feature extractor and projection head.
- We elaborate on the training dynamics and frameworks workflow by diving into the mathematical details of the methods.
- We provide the training report of the methods with top-1/top-5 similarity accuracy for simCLR and redundancy loss for BarlowTwins alongside other training metrics.
- We empirically show how simCLR and BarlowTwins perform compared to the supervised baseline and study the impact of number of labeled samples per class and demonstrate the advantage of SSL.

## 2. Methodologies

In this section, we describe the three methods we implement and evaluate. All three methods share the same convolutional backbone that acts as a feature extractor, denoted as  $f(\cdot)$  throughout in the paper.

### 2.1. SimCLR

SimCLR (Chen et al., 2020a;b) is a contrastive self-supervised framework built around instance discrimination. Given a mini-batch of  $N$  images, two random augmentations are sampled from a predefined distribution  $\mathcal{T}$  and independently applied to each image, yielding  $2N$  augmented views. The two views originating from the same image form a positive pair, while all other  $2N - 2$  views in the batch are treated as negatives. A base encoder  $f(\cdot)$ , parameterized as a convolutional network, maps each view  $\tilde{x}$  to a representation  $h = f(\tilde{x})$ . A projection head  $g(\cdot)$ , implemented as a two-layer MLP with a ReLU non-linearity, further maps  $h$  to

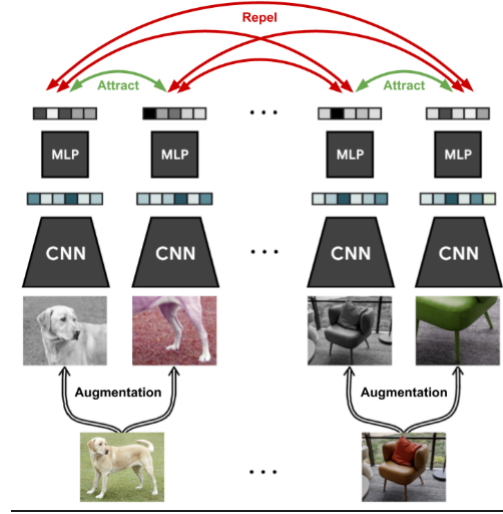


Figure 2. SimCLR framework. Two augmentations of the same image produce two views, which are encoded by a shared feature extractor  $f(\cdot)$  and mapped to embedding by  $g(\cdot)$ . The contrastive loss maximizes agreement between  $z_i$  and  $z_j$  while pushing them apart from embeddings of other images in the batch. After pre-training, the feature extractor learns structural representations.

an embedding  $z = g(h)$  which is further used for computing contrastive loss. After pretraining, the projection head  $g(\cdot)$  is discarded and only the representation  $h$  produced by  $f(\cdot)$  is retained for downstream use, since features taken from before the projection head have consistently been shown to transfer better than those taken after it (Chen et al., 2020a). A schematic of the framework is depicted in Fig. 2.

The contrastive objective used by SimCLR is the InfoNCE loss (Oord et al., 2018). Let  $\text{sim}(u, v) = u^\top v / (\|u\| \|v\|)$  denote the cosine similarity between two vectors, and let  $\tau > 0$  denote a temperature hyperparameter. For a positive pair of indices  $(i, j)$  in a batch of  $2N$  embeddings, the loss takes the form

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (1)$$

where the total loss is computed by averaging  $\ell_{i,j}$  over all positive pairs in the batch. Since cosine similarity is bounded in  $[-1, 1]$ , the temperature is essential to prevent the loss from saturating.

Given Eq. (1), there are two vital properties that should be considered. First, the contrastive loss is implicitly a softmax classification with  $2N - 1$  classes which allows it to be computed efficiently using a single cross-entropy call over the similarity matrix computed using dot product. Second, since the negative samples are drawn entirely from within the batch, the quality of the learned representations is dependent on the batch size: larger batches provide a richer set of negatives and typically yield stronger features. In

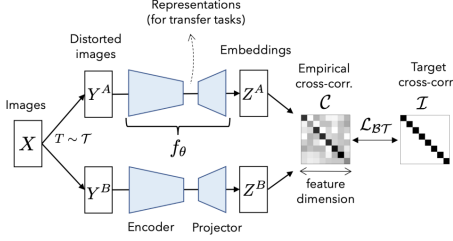


Figure 3. Barlow Twins framework. Two augmented views are passed through the feature extractor  $f(\cdot)$  and projector  $g(\cdot)$  to yield embedding matrices  $Z^A$  and  $Z^B$ . The cross-correlation matrix  $C$  computed between the two batches of embeddings is pushed towards the identity matrix, enforcing uncorrelated views except the same pairs.

our implementation, we adopt the batch size and temperature and training hyperparameters from SimCLRv2 (Chen et al., 2020b) and we use a projection MLP whose hidden dimension is four times the output dimension.

## 2.2. Barlow Twins

While SimCLR explicitly contrasts positive pairs against negatives, Barlow Twins (Zbontar et al., 2021) approaches the problem from a different perspective. Rather than comparing individual sample pairs, it operates directly on the statistics of the embeddings across a batch. Given two batches of embeddings  $Z^A, Z^B \in \mathbb{R}^{B \times D}$  corresponding to the two augmented views of the same  $B$  images, Barlow Twins first normalizes each batch to zero mean and unit variance, and then computes the cross-correlation matrix

$$C_{ij} = \frac{\sum_{b=1}^B \tilde{Z}_{b,i}^A \tilde{Z}_{b,j}^B}{B}, \quad (2)$$

where  $\tilde{Z}^A$  and  $\tilde{Z}^B$  denote the normalized embeddings. The learning objective consists of two terms derived directly from  $C$ :

$$\mathcal{L}_{BT} = \underbrace{\sum_i (1 - C_{ii})^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} C_{ij}^2}_{\text{redundancy reduction term}}, \quad (3)$$

where  $\lambda > 0$  is a positive hyperparameter that balances the two terms. The main intuition of this method is to utilize invariance term to drives the diagonal of  $C$  toward one, which is equivalent to requiring each feature dimension to agree between the two views of the same image and use the redundancy reduction term to drives the off-diagonal entries toward zero, enabling different feature dimensions to encode distinct and uncorrelated aspects of the input. Put together, these two terms push  $C$  towards the identity matrix. Fig. 3 illustrates the workflow and idea of BarlowTwins.

## 2.3. Supervised Baseline

To contextualize the quality of the self-supervised representations, we include a fully supervised baseline in which the same feature extractor  $f(\cdot)$  is trained on the labeled training split of STL-10 for the classification task. In this setting, the projection head is replaced by a linear classifier that maps the features directly to class logits, and the entire network is optimized via the standard cross-entropy loss

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{n=1}^N \sum_{c=1}^C y_{n,c} \log \hat{p}_{n,c}, \quad (4)$$

where  $y_{n,c}$  is the one-hot label,  $\hat{p}_{n,c}$  is the softmax prediction for class  $c$ ,  $N$  is the batch size, and  $C$  is the number of classes.

## 3. Experimental Results

### 3.1. Simulation Setup

**Dataset.** We conduct all experiments on the STL-10 dataset (Chen et al., 2020a), which contains images of 10 classes drawn from a broader ImageNet-like domain. The dataset provides three splits: a small labeled training split of 5,000 images (500 per class), a labeled test split of 8,000 images (800 per class), and a large unlabeled split of 100,000 images sampled from a wider distribution of related categories. The native resolution is  $96 \times 96$ . To reduce computational cost, we downscale all images to  $32 \times 32$ . Self-supervised pretraining of SimCLR and Barlow Twins is performed exclusively on the unlabeled split, while the labeled splits are used for linear probing and for training the supervised baseline.

**Augmentations.** For both SSL methods, we follow the SimCLR augmentation recipe: random horizontal flip, random resized crop to  $32 \times 32$ , color jitter with reduced strength (brightness, contrast, saturation = 0.5, hue = 0.1), random grayscale with probability 0.2, and Gaussian blur with probability 0.5.

**Feature Extractor.** The backbone  $f(\cdot)$  is a small convolutional network consisting of five convolutional blocks with SiLU activations, each followed by batch normalization, with three of them using stride 2 to progressively halve the spatial resolution from  $32 \times 32$  down to  $4 \times 4$ , culminating in a final linear projection to the representation dimension. This architecture is intentionally lightweight to keep the compute budget manageable while remaining representative of a small CNN. For the self-supervised methods, the channel width is  $c_{\text{hid}} = 32$  but for the supervised baseline, we increase it to  $c_{\text{hid}} = 48$  to compensate for the smaller amount of training data. The projection head  $g(\cdot)$  is a two-layer MLP with hidden dimension  $4 \times d$  and output

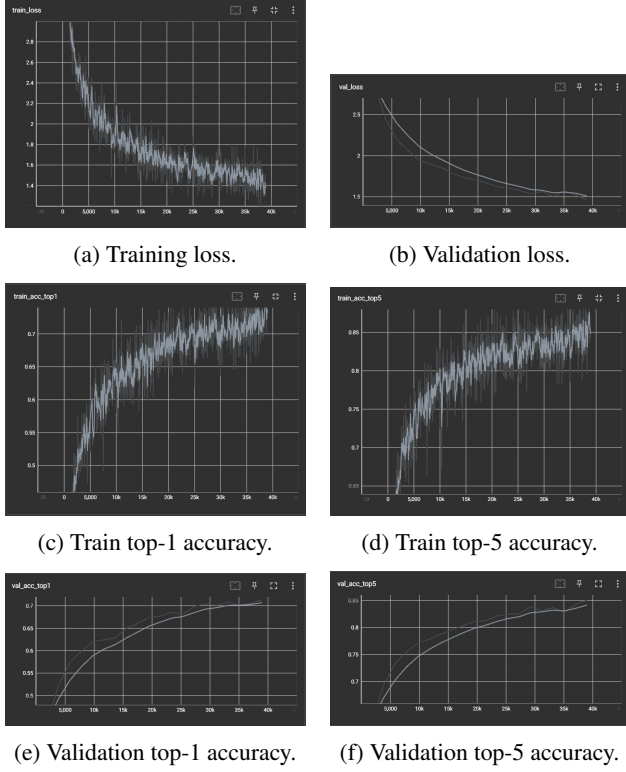


Figure 4. SimCLR training dynamics over 100 epochs on the unlabeled split of STL-10.

dimension  $d = 128$ . For Barlow Twins, a batch normalization layer is inserted into the hidden layer according to the architecture adopted in (Zbontar et al., 2021).

**Training.** SimCLR and Barlow Twins are pretrained for 100 epochs on the unlabeled split with batch size 256, using the AdamW optimizer with a cosine learning rate schedule (initial learning rate  $5 \times 10^{-4}$ , final  $10^{-5}$ ), weight decay  $10^{-4}$ , and  $\tau = 0.07$  for both SimCLR and Barlow Twins. For downstream evaluation, a linear classifier (logistic regression on the frozen  $d$ -dimensional features) is trained on subsets of the labeled training split corresponding to 10, 20, 50, 100, 200, and 500 images per class, and evaluated on the full 8000-image test split. The supervised baseline is trained for 10 epochs on the full labeled training split with batch size 256 and the same optimizer.

### 3.2. SimCLR Training

Fig. 4 shows the training and validation behavior of SimCLR over pretraining. Since the InfoNCE loss itself is not directly interpretable, we additionally monitor two discriminative metrics: the top-1 and top-5 accuracies with which the positive pair for a given anchor is ranked against all other samples in the batch.

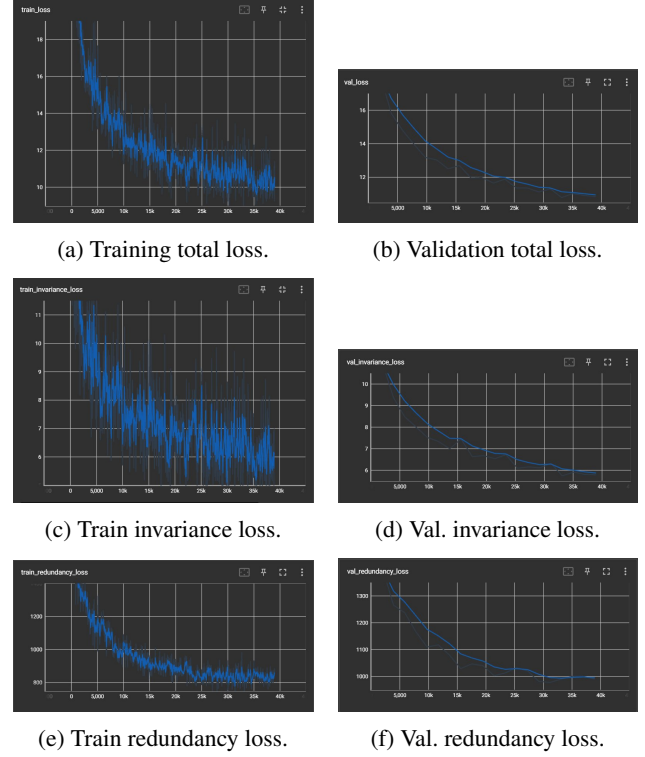


Figure 5. Barlow Twins training dynamics over 100 epochs on the unlabeled split of STL-10.

The training and validation losses in Fig. 4(a)–(b) decrease smoothly throughout training and show no sign of overfitting. The validation curve tracks the training curve closely over the entire run. The top-1 and top-5 similarity accuracies in figures (c)–(f) grow correspondingly toward the 0.7 – 0.8 range, indicating that by the end of training the model can identify the matching positive within a batch of 256 candidates. We also observe that the gap between top-1 and top-5 narrows over training. This shows that later in training, the positive is always among the top few. This behavior is consistent with reports in the literature and suggests that the model is learning an increasingly tight clustering of augmented views generated from the main image. The slope of the loss curve near the end of training indicates that the model converges after pre-training.

### 3.3. Barlow Twins Training

Fig. 5 shows the corresponding curves for Barlow Twins. In addition to the total loss, we separately log its two components defined in Eq. (3): the invariance term, which measures deviation of the diagonal of  $\mathcal{C}$  from one, and the redundancy reduction term, which measures the squared off-diagonal entries.

Both the invariance and redundancy terms decrease over training, indicating that the model is learning, simultane-



Table 1. Classification accuracy on the STL-10 test split over number of labeled images.

| Images/class | SimCLR (%) | Barlow Twins (%) | Supervised (%) |
|--------------|------------|------------------|----------------|
| 10           | 38.54      | 40.70            | —              |
| 20           | 45.30      | 46.58            | —              |
| 50           | 52.61      | 51.70            | —              |
| 100          | 56.09      | 54.46            | —              |
| 200          | 59.59      | 56.98            | —              |
| 500          | 62.49      | 59.46            | 60.35          |

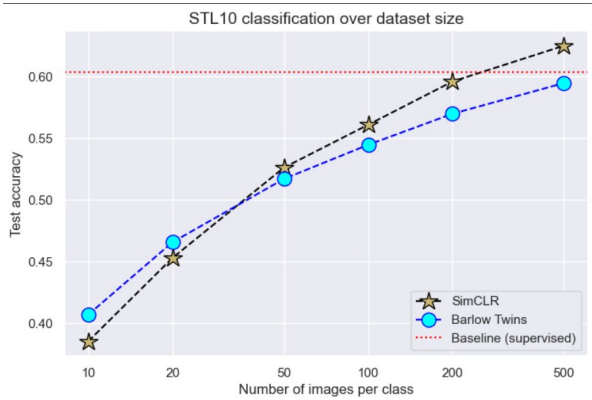


Figure 6. STL-10 test accuracy of linear probes trained on SimCLR and Barlow Twins features with the test accuracy of Supervised method under 500 samples per class.

ously pushing the diagonal of  $\mathcal{C}$  toward one and the off-diagonal entries toward zero. The validation total loss closely follows the training total loss throughout the run, again indicating no overfitting. By the end of training, the curves have flattened, illustrating the convergence.

### 3.4. Downstream Evaluation: STL-10 Classification

To evaluate the quality of the learned representations, we train a linear classifier on features extracted from the frozen backbone and report test accuracy on the STL-10 test split. To characterize the impact of number of labeled samples for each method, we vary the number of labeled images per class from 10 to 500. For comparison, we also report the performance of the supervised baseline, which is trained on the full 500-images-per-class labeled training split and corresponds to a single horizontal line in the plot.

Table 1 reports the numerical results and Fig. 6 visualizes them jointly with the supervised baseline. First, both self-supervised methods produce representations that are clearly useful for classification. Even with only 10 labeled images per class, both linear probes reach roughly 40% test accuracy, indicating that the frozen features already carry a significant amount of class-relevant structure. Second, the two methods exhibit noticeably different behavior as the

label budget increases. Barlow Twins is slightly stronger in the very low-label regime (10 to 20 images per class). As more labels become available, SimCLR outperforms Barlow Twins, reaching 62.49% at 500 images per class compared to 59.46% for Barlow Twins.

Third, the comparison to the supervised baseline is particularly important. The baseline reaches 60.35% test accuracy, which SimCLR matches at 200 images per class (59.59%) and clearly exceeds at 500 (62.49%). In other words, a linear classifier on frozen SimCLR features outperforms the supervised network trained on the same labels, because SimCLR additionally benefits from the 100000 unlabeled images. Barlow Twins comes close to the baseline (59.46% at 500 images per class) but does not overtake it. All curves follow a clear logarithmic trend: accuracy jumps are largest at low label counts (e.g., +7 points for SimCLR going from 10 to 20 images per class) and smaller improvements as labels are added.

These results confirm two of the canonical claims of the SSL literature: (i) self-supervised pretraining on unlabeled data produces features that can be beneficial and more useful compared to supervised learning, and (ii) the advantage of SSL over supervised learning is mostly visible in the low-label practical scenarios, where the supervised network is forced to fit a high-capacity model to very few examples, causing overfitting.

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## 4. Conclusion

Self-supervised learning trains a feature extractor on unlabeled data by solving a pretext task, producing representations that transfer to downstream tasks with few labels. In this project, we implemented and compared two such methods on STL-10: SimCLR, which uses a contrastive InfoNCE objective, and Barlow Twins, which decorrelates features via a cross-correlation loss. Both methods produced features that match or exceed a fully supervised baseline trained on the same labeled set, with the advantage of more effectiveness in low-label regimes. This confirms the core promise of self-supervised learning: effectively learning the structural representations enables effective classification with reduced cost of labeling.

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